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(57) Abstract:

ABSTRACT ADAPTIVE LIGHTING SYSTEM WITH COMPUTER VISION-BASED FOCUS DETECTION An artificial intelligence-driven adaptive lighting system (100) comprises a computer vision module configured to capture real-time video frames of a user via a camera (102), process the video frames using computer vision frameworks (104), and detect facial landmarks, gaze direction and head orientation (106). The system comprises an environmental sensing module having a light dependent resistor sensor connected to a microcontroller configured to measure ambient light intensity (108). The system comprises an artificial intelligence decision module configured to receive behavioral and environmental data (302), analyze the data using focus thresholds and machine learning-based decision logic (304), and classify a user focus level into a high focus state, a medium focus state, or a low focus state (306). The system comprises an Internet of Things-enabled lighting control module configured to adjust brightness and color temperature of a LED lighting system based on the classified focus level.

FORM-2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE

SPECIFICATION

(See section 10 and rule 13)

TITLE OF INVENTION

ADAPTIVE LIGHTING SYSTEM WITH COMPUTER VISION-BASED FOCUS DETECTION

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

[0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to intelligent environmental control systems, and more particularly to an artificial intelligence-driven adaptive lighting system that integrates computer vision-based focus detection, ambient light sensing and Internet of Things-enabled lighting control to dynamically regulate indoor illumination conditions based on user behavioral cues and environmental parameters.

BACKGROUND

[0004] The following text of background art is provided for purposes of understanding and is not an admission that any cited document or technique forms part of the prior art under applicable law.

[0005] Lighting conditions play a role in influencing human concentration, visual comfort and productivity, particularly in environments where individuals engage in prolonged study or work activities. Conventional lighting systems used in homes, offices and educational spaces typically rely on fixed brightness levels, manual adjustments, or simple timer-based scheduling. Although these approaches provide basic illumination, they do not dynamically respond to the user's cognitive state, attention level, or fatigue conditions during extended study sessions.

[0006] Several existing smart lighting solutions have attempted to improve lighting control through environmental sensing or occupancy detection. For example, an intelligent automatic lighting system based on deep learning and computer vision has been disclosed that utilizes deep learning-based cameras to detect human presence and estimate occupant density within a room. Based on parameters such as room size, movement, occupancy and brightness levels, such a system dynamically adjusts artificial lighting in coordination with natural light sources. The system employs luminance sensors or image-based analysis to measure light distribution across different zones and distinguish between natural and artificial light. A central control

unit processes this information in real time and regulates lighting accordingly to reduce energy consumption and provide adaptive illumination. However, such approaches primarily focus on energy efficiency and occupancy-based lighting control by detecting the presence and density of users, without considering their cognitive state or level of attention.

[0007] Other approaches have explored gaze-based lighting control methods. A lighting control method, system, medium and electronic device has been disclosed that enables automatic adjustment of lighting based on user eye movement and intention recognition. Such a system acquires eye movement data such as pupil position, gaze direction, blinking frequency and gaze duration, along with ambient light parameters and lighting parameters. Based on this data, the system identifies the user's intention and accordingly adjusts lighting conditions. The system processes eye movement data to determine whether the user is looking at a specific light source and interprets gaze patterns such as fixed gaze, scanning or repeated viewing between light sources. However, such approaches primarily focus on recognizing user intention based on eye movement to directly control specific light sources, treating gaze as a command input mechanism rather than estimating the user's cognitive state such as attention, focus and fatigue.

[0008] Similarly, a lighting control method, device and electronic equipment used in a display cabinet has been disclosed that acquires eye image data of a user, typically using infrared imaging, and processes this data to determine the user's attention point based on parameters such as pupil position, pupil size and corneal reflection. The system identifies a specific object or unit cabinet that the user is focusing on and selects a corresponding lighting device associated with the identified unit cabinet and adjusts its lighting intensity using a preset lighting strategy. However, such approaches are primarily designed for display environments and focus on identifying the object or area a user is looking at in order to highlight it using predefined lighting strategies, without considering the user's cognitive state or level of concentration.

[0009] Some research works have also explored bio-signal-based lighting control methods that utilize physiological signals such as heart rate variability or electroencephalography to optimize lighting conditions for cognitive performance. However, these approaches generally require wearable monitoring devices or

specialized physiological sensors, which increase system complexity and reduce practicality for everyday usage. Moreover, many current adaptive lighting systems lack the ability to monitor behavioral cues such as gaze direction, facial orientation and user attention in real time without requiring wearable devices.

[0010] Therefore, improved systems and methods that can dynamically adapt lighting conditions based on user cognitive state and behavioral cues without requiring wearable physiological sensors are desired.

[0011] The present invention addresses the above shortcomings of the prior art. However, the invention is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0012] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0013] According to an aspect of the present disclosure, an artificial intelligence-driven adaptive lighting system is provided. The system includes a computer vision module configured to capture real-time video frames of a user via a webcam and process the frames using computer vision frameworks to detect facial landmarks, gaze direction and head orientation. The system includes an environmental sensing module having a light dependent resistor sensor connected to a microcontroller configured to measure ambient light intensity within a study environment. The system includes an artificial intelligence decision module configured to receive behavioral data from the computer vision module and environmental data from the environmental sensing module, analyze the inputs using focus thresholds and machine learning-based decision logic, and classify the user focus level into a high focus state, a medium focus state, or a low focus state. The system includes an Internet of Things-enabled lighting control module having a programmable LED lighting system connected to the microcontroller, wherein the lighting control module is configured to receive lighting parameters from

the artificial intelligence decision module and adjust brightness and color temperature of the LED lighting system based on the classified user focus level.

[0014] This configuration enables non-intrusive monitoring of user cognitive state without requiring wearable physiological sensors, thereby reducing system complexity and improving practicality for everyday usage.

[0015] According to other aspects, the computer vision module may monitor behavioral indicators including eye blink rate, gaze direction, head orientation and facial posture to estimate user attention levels. The monitoring of multiple behavioral indicators allows for more robust focus estimation compared to systems relying on a single behavioral cue.

[0016] According to other aspects, the artificial intelligence decision module may maintain steady medium brightness with cool white light when the user focus level is classified as high focus, apply balanced brightness with neutral white light when classified as medium focus, and increase brightness and shift to a warm light tone when classified as low focus or fatigue. The differentiated lighting responses enable appropriate visual environments tailored to the user's cognitive condition.

[0017] According to other aspects, the computer vision module may generate a no face detected alert when no face is detected in the video frames. This capability helps maintain user engagement during study sessions by providing timely notifications when attention loss is detected.

[0018] According to other aspects, the lighting control module may transmit control signals to the LED lighting system via a serial communication interface. The serial communication interface enables reliable and low-latency transmission of control commands between the processing system and the lighting hardware.

[0019] According to other aspects, the system may automatically increase LED brightness when ambient light levels decrease or when the user focus level drops. This ensures appropriate illumination is maintained throughout study sessions without requiring manual intervention.

[0020] According to other aspects, the system may include a visualization and monitoring module configured to provide a dashboard interface that displays focus score, ambient light percentage, camera brightness, session duration and system alerts. The dashboard interface may display a deep focus mode indicator when the user is in

a high focus state, provide motivational feedback when the user is in a moderate focus state, and generate a low focus alert when attention loss is detected. The real-time visualization enables users to observe how the system monitors their behavioral state and environmental conditions, enhancing transparency and user trust.

[0021] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

[0022] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

[0023] FIG. 1 illustrates a flowchart of a method for overall operation of an AI-driven adaptive smart lighting system, according to aspects of the present disclosure.

[0024] FIG. 2 illustrates a flowchart of a method for computer vision-based focus detection, according to aspects of the present disclosure.

[0025] FIG. 3 illustrates a flowchart of a method for AI decision module operation and lighting parameter determination, according to an embodiment.

[0026] FIG. 4 illustrates a flowchart of a method for adaptive feedback, alert generation and dashboard visualization, according to aspects of the present disclosure.

[0027] FIG. 5 illustrates a real-time monitoring dashboard interface displaying live system metrics, according to an embodiment.

[0028] FIG. 6 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a focus score variation, according to aspects of the present disclosure.

[0029] FIG. 7 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying detection of a sudden focus drop event, according to an embodiment.

[0030] FIG. 8 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying fluctuations in room light and focus score, according to aspects of the present disclosure.

[0031] FIG. 9 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a detected loss of user attention, according to an embodiment.

[0032] FIG. 10 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying variations in system metrics over time, according to aspects of the present disclosure.

[0033] FIG. 11 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying system feedback messages during a study session, according to an embodiment.

[0034] FIG. 12 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying motivational feedback and lighting adjustment notifications, according to aspects of the present disclosure.

[0035] FIG. 13 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a no face detected alert, according to an embodiment.

[0036] FIG. 14 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a distraction warning and lighting adjustment feedback, according to aspects of the present disclosure.

[0037] FIG. 15 illustrates a graph depicting adaptive lighting behavior based on user focus and environmental conditions over time, according to an embodiment.

DETAILED DESCRIPTION

[0038] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

[0039] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements. Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0040] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope

of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0041] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0042] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0043] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0044] FIG. 1 illustrates a flowchart of a method 100 for overall operation of an AI-driven adaptive smart lighting system according to various embodiments. The method 100 describes the operational flow of an integrated system that combines computer vision-based behavioral monitoring, environmental sensing, artificial intelligence-based decision making and Internet of Things (IoT)-enabled lighting control to dynamically adjust indoor illumination conditions based on user behavioral cues and environmental parameters.

[0045] The method 100 includes a step 102 of capturing real-time video frames of a user via a webcam. In step 102, a webcam or camera device captures continuous video frames of a user during a study or work session. The captured video frames provide visual data representing the user's facial features, head position and behavioral patterns. The webcam may be positioned to capture the user's face and upper body

region during the study session. The video frames captured in step 102 serve as input data for subsequent behavioral analysis performed by a computer vision module.

[0046] The method 100 includes a step 104 of processing the frames using computer vision frameworks. In step 104, the captured video frames are processed using computer vision frameworks such as OpenCV and MediaPipe. These frameworks enable the system to analyze the visual content of each frame and extract relevant facial features with high accuracy. The processing performed in step 104 prepares the video data for facial landmark detection and behavioral indicator extraction. OpenCV provides image processing capabilities, while MediaPipe provides facial landmark detection functionality.

[0047] The method 100 includes a step 106 of detecting facial landmarks, gaze direction and head orientation. In step 106, the computer vision module detects facial landmarks from the processed video frames and determines the user's gaze direction and head orientation. Facial landmark detection involves identifying specific points on the user's face, such as eye positions, nose position and mouth position. Based on the detected facial landmarks, the system determines where the user is looking (gaze direction) and how the user's head is positioned (head orientation). The behavioral indicators extracted in step 106 provide information about the user's attention state during the study session.

[0048] The method 100 includes a step 108 of measuring ambient light intensity using a light dependent resistor (LDR) sensor via a microcontroller. In step 108, an environmental sensing module measures the ambient light intensity present in the study environment. The LDR sensor detects variations in surrounding light levels and converts the detected light intensity into electrical signals. These electrical signals are read by a microcontroller, such as an Arduino microcontroller, and transmitted to a central processing system. The ambient light data collected in step 108 provides environmental context for the lighting control decisions made by the system.

[0049] The method 100 includes a step 110 of analyzing the behavioral and environmental data in an AI decision module. In step 110, an artificial intelligence decision module receives the behavioral data extracted from the computer vision module and the environmental data obtained from the sensing module. The AI decision module analyzes these inputs using focus thresholds and machine learning-based

decision logic. The analysis performed in step 110 evaluates the user's cognitive focus level based on behavioral indicators such as eye blink rate, gaze direction, head orientation and facial posture, in combination with the current ambient lighting conditions.

[0050] The method 100 includes a step 112 of classifying the user focus level. In step 112, the AI decision module classifies the user's focus level into one of several focus states based on the analysis performed in step 110. The focus states may include a high focus state, a medium focus state, or a low focus state (which may also indicate fatigue). The classification performed in step 112 determines the appropriate lighting configuration for the current user state. For example, when stable gaze direction and minimal head movement are observed, the system may classify the user as being in a high focus state. When reduced attention, increased blinking, or irregular head movement is detected, the system may classify the user as being in a medium focus state or a low focus state.

[0051] The method 100 includes a step 114 of transmitting control signals to an IoT-enabled LED lighting system to adjust brightness and color temperature. In step 114, the AI decision module transmits lighting parameters to an IoT-enabled lighting control module based on the classified user focus level. The lighting control module receives the control signals via a serial communication interface and adjusts the brightness and color temperature of a programmable LED lighting system, such as a NeoPixel programmable LED module. When the user focus level is classified as high focus, the system may maintain steady medium brightness with cool white light. When the user focus level is classified as medium focus, the system may apply balanced brightness with neutral white light. When the user focus level is classified as low focus or fatigue, the system may increase brightness and shift to a warm light tone.

[0052] The method 100 includes a step 116 of displaying real-time metrics on a monitoring dashboard. In step 116, a visualization and monitoring module displays system parameters and operational status on a dashboard interface. The dashboard may present information such as the user's focus score, ambient light percentage, camera brightness level, session duration and system alerts or feedback messages. The real-time display provided in step 116 enables users to observe how the system monitors

their behavioral state and environmental conditions. The dashboard interface may be implemented using a web-based interface framework.

[0053] The method 100 includes a step 118 of continuing the adaptive monitoring loop. In step 118, the system returns to step 102 to continue capturing video frames and monitoring the user's behavioral state and environmental conditions. The continuous monitoring loop enables the system to dynamically adapt lighting conditions in response to changes in user attention levels and ambient illumination throughout the study session. Through the continuous interaction between behavioral monitoring, environmental sensing, intelligent decision-making and adaptive lighting control, the system maintains a responsive lighting environment capable of adapting in real time to the user's cognitive state and surrounding illumination conditions.

[0054] The technical architecture of the method 100 involves data flow between multiple functional modules. The computer vision module captures and processes video frames to extract behavioral data, which is forwarded to the AI decision module. The environmental sensing module collects ambient light data via the LDR sensor and microcontroller, which is also forwarded to the AI decision module. The AI decision module analyzes both data streams and generates lighting control commands, which are transmitted to the IoT lighting control module. The IoT lighting control module adjusts the LED lighting parameters based on the received commands. The visualization and monitoring module receives system status information and displays real-time metrics to the user. This modular architecture enables coordinated operation of behavioral monitoring, environmental sensing, intelligent decision-making and adaptive lighting control within a unified framework.

[0055] FIG. 2 illustrates a flowchart of a method 200 for computer vision-based focus detection according to various embodiments. The method 200 describes the operational flow of a computer vision module that captures and processes real-time video frames to detect facial landmarks, monitor behavioral indicators and estimate user attention levels during study or work sessions. The method 200 operates as a component of the AI-driven adaptive smart lighting system and provides behavioral data to an artificial intelligence decision module for lighting parameter determination.

[0056] The method 200 includes a step 202 of capturing real-time video frames from a webcam during a study session. In step 202, a webcam or camera device

captures continuous video frames of a user while the user engages in study or work activities. The webcam may be positioned to capture the user's face and upper body region during the study session. The captured video frames provide visual data representing the user's facial features, eye positions, head position and behavioral patterns. The video frames captured in step 202 serve as input data for subsequent facial landmark detection and behavioral analysis. The frame capture rate may be configured to provide sufficient temporal resolution for detecting changes in user behavior and attention state.

[0057] The method 200 includes a step 204 of processing the frames using computer vision frameworks for facial landmark detection. In step 204, the captured video frames are processed using computer vision libraries such as OpenCV and MediaPipe. OpenCV provides image processing capabilities including frame acquisition, color space conversion, image filtering and feature extraction. MediaPipe provides facial landmark detection functionality that enables identification of specific facial feature points with high accuracy. The processing performed in step 204 prepares the video data for extraction of behavioral indicators by detecting and localizing facial features within each frame. The facial landmark detection algorithm may identify multiple landmark points on the user's face, including points corresponding to eye positions, eyebrow positions, nose position, mouth position and facial contour. These landmark points provide geometric information about the user's facial structure and orientation within the captured frame.

[0058] The method 200 includes a step 206 of determining whether a face is detected in the video frames. In step 206, the computer vision module evaluates the output of the facial landmark detection algorithm to determine whether a valid face is present in the current video frame. The face detection determination may be based on the presence and confidence level of detected facial landmarks. When the

[0059] FIG. 2 illustrates a flowchart of a method 200 for computer vision-based focus detection according to various embodiments. The method 200 describes the operational flow of a computer vision module that captures and processes real-time video frames to detect facial landmarks, monitor behavioral indicators and estimate user attention levels during study or work sessions. The method 200 operates as a component within the AI-driven adaptive smart lighting system and provides

behavioral data to an artificial intelligence decision module for lighting parameter determination.

[0060] The method 200 includes a step 202 of capturing real-time video frames from a webcam during a study session. In step 202, a webcam or camera device captures continuous video frames of a user while the user engages in study or work activities. The webcam may be positioned to capture the user's face and upper body region during the study session. The captured video frames provide visual data representing the user's facial features, eye positions, head position and behavioral patterns. The video frames captured in step 202 serve as input data for subsequent facial landmark detection and behavioral analysis operations. The frame capture rate may be configured to provide sufficient temporal resolution for detecting changes in user behavior and attention state.

[0061] The method 200 includes a step 204 of processing the video frames using computer vision frameworks for facial landmark detection. In step 204, the captured video frames are processed using computer vision libraries such as OpenCV and MediaPipe. OpenCV provides image processing capabilities including frame acquisition, color space conversion, image filtering and feature extraction operations. MediaPipe provides facial landmark detection functionality that enables identification of specific facial feature points with high accuracy. The processing performed in step 204 involves applying facial landmark detection algorithms to each video frame to identify and locate facial feature points. The facial landmark detection algorithms may identify multiple landmark points on the user's face, including points corresponding to eye positions, eyebrow positions, nose position, mouth position and facial contour positions. The detected landmark points provide geometric information about the user's facial structure and orientation within each video frame.

[0062] The method 200 includes a step 206 of determining whether a face is detected in the video frames. In step 206, the computer vision module evaluates the output of the facial landmark detection algorithms to determine whether a valid face has been detected in the current video frame. The face detection determination may be based on whether a sufficient number of facial landmark points have been successfully identified and whether the detected landmark points form a valid facial geometry. When a face is detected, the method 200 proceeds to step 210 for behavioral indicator

monitoring. When no face is detected, the method 200 proceeds to step 208 for alert generation.

[0063] The method 200 includes a step 208 of generating a no face detected alert and requesting the user to return to the screen when no face is detected. In step 208, the computer vision module generates an alert notification indicating that no face has been detected in the video frames. The alert may be displayed on a monitoring dashboard interface to inform the user that the user's face is not visible to the webcam. The alert message may request the user to return to the screen or adjust the user's position so that the webcam can capture the user's face. The no face detected condition may indicate that the user has moved away from the study area, turned away from the screen, or is otherwise not positioned within the webcam's field of view. After generating the alert in step 208, the method 200 may proceed to step 218 to continue the monitoring loop.

[0064] The method 200 includes a step 210 of monitoring eye blink rate, gaze direction, head orientation and facial posture when a face is detected. In step 210, the computer vision module analyzes the detected facial landmarks to extract behavioral indicators that provide information about the user's attention state. Eye blink rate monitoring involves tracking the positions of eyelid landmarks over time to detect blink events and calculate the frequency of blinking. Gaze direction estimation involves analyzing the positions of eye landmarks, including pupil positions and eye corner positions, to determine where the user is looking. Head orientation tracking involves analyzing the three-dimensional orientation of the face based on the relative positions of facial landmarks to determine whether the user's head is facing forward, tilted, or turned away from the screen. Facial posture analysis involves evaluating the overall positioning and alignment of facial features to assess the user's engagement level. The behavioral indicators extracted in step 210 provide information about whether the user is actively focused on the study task, distracted, or experiencing fatigue. For example, frequent blinking may indicate tiredness, downward gaze may indicate reduced attention, and irregular head movement may indicate distraction.

[0065] The method 200 includes a step 212 of evaluating the behavioral indicators against focus thresholds for attention estimation. In step 212, the computer vision module compares the extracted behavioral indicators against predefined focus thresholds to estimate the user's attention level. The focus thresholds may define

acceptable ranges or values for each behavioral indicator that correspond to different attention states. For example, a blink rate threshold may distinguish between normal blinking associated with focused attention and elevated blinking associated with fatigue. A gaze direction threshold may distinguish between forward-facing gaze associated with screen engagement and off-screen gaze associated with distraction. A head orientation threshold may distinguish between stable head positioning associated with concentration and frequent head movement associated with reduced attention. The evaluation performed in step 212 combines the assessments of multiple behavioral indicators to generate an overall attention estimation for the user.

[0066] The method 200 includes a step 214 of converting the behavioral features into structured data representing the user state. In step 214, the computer vision module converts the extracted behavioral indicators and attention estimation results into a structured data format suitable for transmission to the artificial intelligence decision module. The structured data may include numerical values representing the current measurements of each behavioral indicator, such as blink rate values, gaze direction coordinates, head orientation angles and facial posture metrics. The structured data may also include computed attention scores or focus level classifications derived from the threshold evaluation performed in step 212. The conversion to structured data format facilitates efficient processing and analysis by downstream system components.

[0067] The method 200 includes a step 216 of forwarding the processed data to the AI decision module for analysis. In step 216, the computer vision module transmits the structured behavioral data to the artificial intelligence decision module. The AI decision module receives the behavioral data along with environmental data from an environmental sensing module and performs further analysis to determine appropriate lighting parameters. The data transmission in step 216 may occur through inter-process communication mechanisms, shared memory structures, or message passing interfaces depending on the system implementation. The forwarding of processed data in step 216 enables the AI decision module to make lighting control decisions based on the user's current behavioral state.

[0068] The method 200 includes a step 218 of returning to the continuous monitoring loop. In step 218, the method 200 returns to step 202 to continue capturing video frames and monitoring the user's behavioral state. The continuous monitoring

loop enables the computer vision module to track changes in user behavior and attention levels over time throughout the study session. By continuously processing video frames and extracting behavioral indicators, the method 200 provides ongoing behavioral data to the AI decision module, enabling the adaptive lighting system to respond dynamically to changes in user attention and engagement. The monitoring loop continues until the study session ends or the system is deactivated.

[0069] The facial landmark extraction algorithms employed in the method 200 may utilize machine learning-based detection models that have been trained on facial image datasets to identify landmark points with high accuracy across varying lighting conditions, user positions and facial expressions. The MediaPipe framework may provide a face mesh model that detects multiple facial landmark points distributed across the face surface, enabling detailed analysis of facial geometry and orientation. The landmark detection process may involve initial face detection to locate the face region within the video frame, followed by landmark localization to identify specific feature points within the detected face region.

[0070] The attention estimation techniques employed in the method 200 may combine multiple behavioral indicators using weighted scoring or classification algorithms to generate focus level assessments. The weighting of different behavioral indicators may be configured based on the relative importance of each indicator for attention estimation. For example, gaze direction may receive higher weighting than blink rate in determining whether the user is engaged with the study task. The attention estimation may also incorporate temporal analysis to detect patterns of behavior over time, such as sustained off-screen gaze or progressive increases in blink rate that may indicate developing fatigue.

[0071] FIG. 3 illustrates a flowchart of a method 300 for AI decision module operation and lighting parameter determination according to various embodiments. The method 300 describes the operational flow of an artificial intelligence decision module that receives behavioral and environmental data, analyzes the data to classify user focus levels, determines appropriate lighting parameters and transmits control commands to an Internet of Things (IoT)-enabled lighting system. The method 300 operates as a component within the AI-driven adaptive smart lighting system and coordinates with

the computer vision module and environmental sensing module to enable dynamic lighting control based on user cognitive state and surrounding illumination conditions.

[0072] The method 300 includes a step 302 of receiving behavioral data from the computer vision module and environmental data from the sensing module. In step 302, the artificial intelligence decision module receives structured behavioral data that has been extracted and processed by the computer vision module. The behavioral data may include measurements of eye blink rate, gaze direction coordinates, head orientation angles and facial posture metrics. The behavioral data may also include computed attention scores or preliminary focus level assessments derived from threshold evaluations performed by the computer vision module. Concurrently, the artificial intelligence decision module receives environmental data from the environmental sensing module. The environmental data may include ambient light intensity measurements obtained from a light dependent resistor (LDR) sensor connected to a microcontroller. The combination of behavioral data and environmental data in step 302 provides the artificial intelligence decision module with information about both the user's cognitive state and the surrounding illumination conditions.

[0073] The method 300 includes a step 304 of analyzing the inputs using focus thresholds and machine learning-based decision logic. In step 304, the artificial intelligence decision module processes the received behavioral data and environmental data to evaluate the user's cognitive focus level. The analysis performed in step 304 may involve comparing behavioral indicators against predefined focus thresholds to assess the user's attention state. For example, the artificial intelligence decision module may evaluate whether the user's blink rate exceeds a fatigue threshold, whether the user's gaze direction indicates screen engagement or distraction, and whether the user's head orientation indicates stable positioning or frequent movement. The analysis may also incorporate machine learning-based decision logic that has been trained to recognize patterns of behavior associated with different focus states. The machine learning-based decision logic may combine multiple behavioral indicators using weighted scoring or classification algorithms to generate focus level assessments. The analysis performed in step 304 may also consider the current ambient light intensity to determine whether the existing lighting environment is suitable for the user's cognitive state.

[0074] The method 300 includes a step 306 of classifying the user focus level into a high focus state, a medium focus state, or a low focus state. In step 306, the artificial intelligence decision module assigns a focus classification to the user based on the analysis performed in step 304. The focus classification may include a high focus state, a medium focus state, or a low focus state (which may also indicate fatigue). The high focus state may be assigned when the user exhibits stable gaze direction, minimal head movement and normal blink rate, indicating active engagement with the study task. The medium focus state may be assigned when the user exhibits moderate variations in gaze direction or head orientation, indicating partial attention or slight distraction. The low focus state may be assigned when the user exhibits frequent off-screen gaze, irregular head movement, elevated blink rate, or other behavioral indicators associated with reduced attention or fatigue. The focus classification determined in step 306 serves as the basis for selecting appropriate lighting parameters in subsequent steps.

[0075] The method 300 includes a step 308 of maintaining steady medium brightness with cool white light for high focus. In step 308, when the user focus level is classified as high focus in step 306, the artificial intelligence decision module determines that the current lighting configuration may be maintained at steady medium brightness with cool white light. Cool white light may support alertness and concentration during periods of high cognitive engagement. The steady medium brightness level may provide sufficient illumination for reading and study activities without causing visual discomfort. By maintaining stable lighting conditions during high focus periods, the system may avoid unnecessary lighting changes that could distract the user from the study task.

[0076] The method 300 includes a step 310 of applying balanced brightness with neutral white light for medium focus. In step 310, when the user focus level is classified as medium focus in step 306, the artificial intelligence decision module determines that balanced brightness with neutral white light may be applied. Neutral white light may provide a comfortable visual environment that supports moderate attention levels. The balanced brightness level may be adjusted based on the current ambient light intensity to maintain appropriate illumination. The lighting configuration applied in step 310 may help the user maintain concentration while providing a comfortable visual environment.

[0077] The method 300 includes a step 312 of increasing brightness and shifting to a warm light tone for low focus or fatigue. In step 312, when the user focus level is classified as low focus or fatigue in step 306, the artificial intelligence decision module determines that brightness may be increased and the color temperature may be shifted to a warm light tone. Increased brightness may help counteract reduced attention by providing enhanced visual stimulation. The warm light tone may reduce eye strain and improve visual comfort during periods of fatigue. The lighting adjustments applied in step 312 may help the user recover concentration and reduce visual fatigue during extended study sessions.

[0078] The method 300 includes a step 314 of transmitting the determined lighting parameters to the IoT lighting control module via a serial communication interface. In step 314, the artificial intelligence decision module transmits control commands containing the determined lighting parameters to an IoT lighting control module. The control commands may specify brightness level, color temperature and overall light intensity values to be applied by the lighting system. The transmission of control commands in step 314 may occur through a serial communication interface that connects the central processing system to a microcontroller. The serial communication interface may enable reliable and low-latency transmission of control commands between the artificial intelligence decision module and the lighting hardware.

[0079] The method 300 includes a step 316 of the microcontroller regulating LED brightness, color temperature and light intensity. In step 316, a microcontroller, such as an Arduino microcontroller, receives the control commands transmitted in step 314 and regulates the parameters of a programmable LED lighting system. The programmable LED lighting system may be a NeoPixel programmable LED module or similar addressable LED system. The microcontroller may adjust the brightness level of the LED lighting system by modifying the power output or duty cycle of the LED driver signals. The microcontroller may adjust the color temperature by controlling the relative intensity of different color channels within the LED lighting system. The microcontroller may also adjust the overall light intensity based on the control commands received from the artificial intelligence decision module. The regulation performed in step 316 enables the lighting system to dynamically adapt the illumination

conditions in real time according to the user's focus state and environmental lighting levels.

[0080] The method 300 includes a step 318 of determining whether conditions have changed. In step 318, the artificial intelligence decision module evaluates whether the user's behavioral state or the environmental lighting conditions have changed since the previous analysis cycle. The determination in step 318 may be based on comparing current behavioral data and environmental data with previously received data to detect changes in user attention level or ambient illumination. When conditions have changed, the method 300 may return to step 302 to receive updated behavioral and environmental data and perform a new analysis cycle. When conditions have not changed, the method 300 may proceed to step 320 to maintain the current lighting configuration.

[0081] The method 300 includes a step 320 of maintaining current lighting conditions for sustained study productivity. In step 320, when the determination in step 318 indicates that conditions have not changed, the artificial intelligence decision module maintains the current lighting parameters without modification. By maintaining stable lighting conditions when the user's focus state and environmental conditions remain consistent, the system may avoid unnecessary lighting fluctuations that could distract the user or cause visual discomfort. The method 300 may then return to step 318 to continue monitoring for condition changes, thereby establishing a continuous monitoring loop that enables the system to respond dynamically to changes in user behavior and environmental conditions throughout the study session.

[0082] The artificial intelligence decision module employed in the method 300 may utilize a combination of rule-based logic and machine learning-based classification to determine appropriate lighting parameters. The rule-based logic may define specific lighting configurations for each focus state based on predefined relationships between focus levels and lighting parameters. The machine learning-based classification may enable the system to adapt to individual user behavior patterns and improve focus estimation accuracy over time. The combination of rule-based logic and machine learning-based classification may provide both systematic evaluation and adaptive learning capabilities.

[0083] The lighting parameter determination performed by the method 300 may consider the relationship between focus level and lighting configuration as follows:

when the focus level is high focus, the lighting brightness may be set to medium steady brightness and the color temperature may be set to cool white light; when the focus level is medium focus, the lighting brightness may be set to balanced brightness and the color temperature may be set to neutral white light; when the focus level is low focus or fatigue, the lighting brightness may be increased and the color temperature may be shifted to warm light tone.

[0084] Working Example 1: Focus Classification and Lighting Adjustment Process

[0085] In a working example demonstrating the focus classification and lighting adjustment process, a user engages in a study session while the AI-driven adaptive smart lighting system monitors the user's behavioral state and environmental conditions. The computer vision module captures real-time video frames of the user and extracts behavioral indicators including eye blink rate, gaze direction, head orientation and facial posture. The environmental sensing module measures the ambient light intensity using an LDR sensor connected to an Arduino microcontroller.

[0086] At the beginning of the study session, the user exhibits stable gaze direction focused on the screen, minimal head movement and a normal blink rate of approximately 15 blinks per minute. The artificial intelligence decision module receives the behavioral data and environmental data in step 302. In step 304, the artificial intelligence decision module analyzes the inputs and determines that the behavioral indicators are within the thresholds associated with high cognitive engagement. In step 306, the artificial intelligence decision module classifies the user focus level as high focus. In step 308, the artificial intelligence decision module determines that steady medium brightness with cool white light may be maintained. In step 314, the artificial intelligence decision module transmits the lighting parameters to the IoT lighting control module. In step 316, the Arduino microcontroller regulates the NeoPixel LED module to maintain medium brightness at approximately 60 percent intensity with a color temperature of approximately 5000 Kelvin (cool white).

[0087] After approximately 25 minutes of study, the user begins to exhibit behavioral changes including increased blink rate of approximately 22 blinks per minute, occasional off-screen gaze and slight head tilting. The computer vision module detects these behavioral changes and forwards updated behavioral data to the artificial

intelligence decision module. In step 318, the artificial intelligence decision module determines that conditions have changed. The method 300 returns to step 302 to receive the updated data. In step 304, the artificial intelligence decision module analyzes the updated behavioral indicators and determines that the user's attention level has decreased. In step 306, the artificial intelligence decision module classifies the user focus level as medium focus. In step 310, the artificial intelligence decision module determines that balanced brightness with neutral white light may be applied. In step 314, the artificial intelligence decision module transmits the updated lighting parameters. In step 316, the Arduino microcontroller adjusts the NeoPixel LED module to balanced brightness at approximately 65 percent intensity with a color temperature of approximately 4000 Kelvin (neutral white).

[0088] After approximately 45 minutes of study, the user exhibits further behavioral changes including elevated blink rate of approximately 28 blinks per minute, frequent downward gaze, irregular head movement and facial posture indicating fatigue. Concurrently, the environmental sensing module detects that the ambient light intensity has decreased as natural daylight diminishes. The artificial intelligence decision module receives the updated behavioral and environmental data. In step 304, the artificial intelligence decision module analyzes the inputs and determines that the behavioral indicators are consistent with reduced attention and fatigue, and that the ambient light level has decreased. In step 306, the artificial intelligence decision module classifies the user focus level as low focus or fatigue. In step 312, the artificial intelligence decision module determines that brightness may be increased and the color temperature may be shifted to a warm light tone. In step 314, the artificial intelligence decision module transmits the updated lighting parameters. In step 316, the Arduino microcontroller adjusts the NeoPixel LED module to increased brightness at approximately 80 percent intensity with a color temperature of approximately 3000 Kelvin (warm white). The increased brightness compensates for both the decreased ambient light and the user's reduced attention level, while the warm light tone reduces eye strain and improves visual comfort during the fatigue period.

[0089] Throughout the study session, the method 300 continues to monitor the user's behavioral state and environmental conditions, dynamically adjusting lighting parameters as conditions change. The continuous monitoring loop established by steps

318 and 320 enables the system to maintain responsive lighting control that adapts to the user's cognitive state and surrounding illumination conditions in real time.

[0090] FIG. 4 illustrates a flowchart of a method 400 for adaptive feedback, alert generation and dashboard visualization according to various embodiments. The method 400 describes the operational flow of a visualization and monitoring module that provides real-time display of system parameters, generates adaptive alerts based on user behavioral state and environmental conditions, and presents feedback messages to support user awareness during study or work sessions. The method 400 operates as a component within the AI-driven adaptive smart lighting system and coordinates with the computer vision module, environmental sensing module and artificial intelligence decision module to provide users with transparency into the adaptive lighting control process.

[0091] The method 400 includes a step 402 of providing a real-time display of system parameters and status via a web-based dashboard interface. In step 402, a visualization and monitoring module presents system information through a dashboard interface that may be implemented using a web-based framework. The dashboard interface may display system status indicators, control elements and graphical visualizations that enable users to observe the current operational state of the adaptive lighting system. The dashboard interface may include a sidebar panel containing system controls such as camera activation buttons and alert enable toggles, along with status indicators showing connectivity states for hardware components such as the microcontroller and webcam. The real-time display provided in step 402 enables users to monitor how the system analyzes behavioral and environmental data and adjusts lighting conditions accordingly.

[0092] The method 400 includes a step 404 of presenting focus score, ambient light percentage, camera brightness, session duration and system alerts. In step 404, the dashboard interface displays multiple system parameters that provide information about the user's behavioral state and environmental conditions. The focus score may represent a computed value indicating the user's current level of cognitive engagement based on behavioral indicators extracted by the computer vision module. The ambient light percentage may represent the current illumination level measured by the light dependent resistor sensor. The camera brightness may represent the brightness level

detected within the captured video frames. The session duration may represent the elapsed time since the study session began. System alerts may include notification messages generated in response to detected events such as attention loss, environmental lighting changes, or focus state transitions. The presentation of these parameters in step 404 enables users to observe how the system monitors their behavioral state and surrounding environmental conditions.

[0093] The method 400 includes a step 406 of determining the current focus state. In step 406, the visualization and monitoring module receives focus classification information from the artificial intelligence decision module and determines the appropriate feedback and display configuration based on the current focus state. The focus state may be classified as a high focus state (which may also be referred to as deep focus), a moderate focus state, or a low focus state. The determination performed in step 406 enables the dashboard interface to provide focus-state-specific feedback and visual indicators to the user.

[0094] The method 400 includes a step 408 of displaying a Deep Focus Mode indicator with stable system metrics for deep or high focus. In step 408, when the focus state is determined to be high focus or deep focus in step 406, the dashboard interface displays a Deep Focus Mode indicator to inform the user that the user is in a highly attentive state. The dashboard may display stable system metrics showing consistent focus score values, room light percentage and camera brightness levels. The Deep Focus Mode indicator may be presented as a notification banner or status message within a system feedback section of the dashboard interface. By displaying the Deep Focus Mode indicator, the system provides positive reinforcement to the user and confirms that the current behavioral state is conducive to productive study activities.

[0095] The method 400 includes a step 410 of providing motivational feedback and monitoring lighting conditions for moderate focus. In step 410, when the focus state is determined to be moderate focus in step 406, the dashboard interface provides motivational feedback messages to encourage the user to maintain concentration. The motivational feedback may include messages such as encouragement to continue studying or acknowledgment of consistent effort. The dashboard interface may continue to display system metrics and monitor environmental lighting conditions while the user is in the moderate focus state. The motivational feedback provided in

step 410 may help the user maintain attention and engagement during periods of partial concentration.

[0096] The method 400 includes a step 412 of generating a No Face or Low Focus alert and encouraging concentration. In step 412, when the focus state is determined to be low focus or when the computer vision module detects that no face is present in the video frames, the dashboard interface generates an alert notification. The No Face Detected alert may request the user to return to the screen or adjust the user's position so that the webcam can capture the user's face. The Low Focus alert may encourage the user to concentrate and avoid distractions. The alert messages may be displayed within a system feedback section of the dashboard interface using visual indicators such as highlighted text or colored notification banners. The alert generation performed in step 412 enables proactive user engagement when attention loss is detected.

[0097] The method 400 includes a step 414 of automatically increasing LED brightness when ambient light decreases or user focus drops. In step 414, the visualization and monitoring module coordinates with the IoT lighting control module to automatically adjust LED brightness in response to changes in ambient light levels or user focus state. When the environmental sensing module detects that ambient light intensity has decreased, the system may increase LED brightness to compensate for the reduced environmental illumination. When the artificial intelligence decision module classifies the user focus level as low focus or fatigue, the system may increase LED brightness to provide enhanced visual stimulation. The automatic brightness adjustment performed in step 414 ensures that appropriate illumination is maintained throughout study sessions without requiring manual intervention. The dashboard interface may display feedback messages indicating that LED brightness has been adjusted in response to environmental or behavioral changes.

[0098] The method 400 includes a step 416 of determining whether ambient light conditions have changed. In step 416, the visualization and monitoring module evaluates whether the ambient light intensity measured by the environmental sensing module has changed since the previous monitoring cycle. The determination in step 416 may be based on comparing current ambient light readings with previously recorded values to detect increases or decreases in environmental illumination. When ambient light conditions have changed, the method 400 may proceed to step 418 to

regulate LED parameters accordingly. When ambient light conditions have not changed, the method 400 may continue monitoring and return to step 406 to evaluate the current focus state.

[0099] The method 400 includes a step 418 of the microcontroller regulating LED parameters including brightness, color temperature and light intensity in real time. In step 418, a microcontroller, such as an Arduino microcontroller, receives control commands from the artificial intelligence decision module and regulates the parameters of a programmable LED lighting system. The microcontroller may adjust the brightness level, color temperature and overall light intensity of the LED lighting system based on the control commands received. The regulation performed in step 418 enables the lighting system to dynamically adapt the illumination conditions in real time according to the user's focus state and environmental lighting levels. The dashboard interface may display updated system metrics reflecting the adjusted lighting parameters.

[0100] The method 400 includes a step 420 of helping users understand system response to their behavioral state and environmental conditions through the interface. In step 420, the dashboard interface provides information and feedback that enables users to understand how the adaptive lighting system responds to changes in their behavioral state and surrounding environmental conditions. The dashboard may display graphical visualizations showing trends in focus score, ambient light percentage and camera brightness over time. The dashboard may also display feedback messages explaining lighting adjustments made by the system, such as notifications indicating that room light is decreasing and LED brightness has been adjusted. By providing this information, the visualization and monitoring module enhances transparency and usability, allowing users to observe how lighting conditions are dynamically adapted to support concentration and a comfortable study environment. The method 400 may then return to step 402 to continue the real-time monitoring and feedback loop.

[0101] The dashboard interface employed in the method 400 may display a graphical visualization showing multiple tracked parameters plotted over time. The graphical visualization may include a line graph with a horizontal axis representing elapsed time and a vertical axis representing percentage values. The line graph may display traces representing room light percentage, camera brightness percentage and

focus score using different colors to distinguish between the parameters. The graphical visualization enables users to observe trends and patterns in their focus levels and environmental conditions over time.

[0102] The feedback messages generated by the method 400 may include multiple types of notifications based on the detected system state. When the user is in a high focus state, the dashboard may display a message indicating that the user is in Deep Focus Mode. When the user is in a moderate focus state, the dashboard may display motivational messages encouraging continued effort. When no face is detected, the dashboard may display an alert requesting the user to return to the screen. When the user exhibits signs of distraction, the dashboard may display a warning message encouraging the user to stay focused and avoid distractions. When ambient light conditions change, the dashboard may display a notification indicating that LED brightness has been adjusted.

[0103] Working Example 2: Alert Generation and Feedback Mechanism

[0104] In a working example demonstrating the alert generation and feedback mechanism, a user engages in a study session while the AI-driven adaptive smart lighting system monitors the user's behavioral state and environmental conditions. The visualization and monitoring module displays system parameters on the dashboard interface, including focus score, ambient light percentage, camera brightness and session duration.

[0105] At the beginning of the study session, the user exhibits stable gaze direction and minimal head movement, and the artificial intelligence decision module classifies the user focus level as high focus. In step 406, the visualization and monitoring module determines that the current focus state is high focus. In step 408, the dashboard interface displays a Deep Focus Mode indicator with a message such as "You are in Deep Focus Mode" within the system feedback section. The graphical visualization displays stable focus score values at approximately 85 percent, room light percentage at approximately 65 percent and camera brightness at approximately 70 percent.

[0106] After approximately 15 minutes of study, the ambient light intensity in the study environment begins to decrease as natural daylight diminishes. In step 416, the visualization and monitoring module determines that ambient light conditions have

changed. In step 414, the system automatically increases LED brightness to compensate for the reduced environmental illumination. In step 418, the Arduino microcontroller regulates the NeoPixel LED module to increase brightness from approximately 55 percent to approximately 70 percent intensity. The dashboard interface displays a feedback message indicating "Room light is decreasing, LED brightness adjusted" within the system feedback section.

[0107] After approximately 25 minutes of study, the user temporarily looks away from the screen and the computer vision module detects reduced gaze engagement. The artificial intelligence decision module classifies the user focus level as moderate focus. In step 406, the visualization and monitoring module determines that the current focus state is moderate focus. In step 410, the dashboard interface provides motivational feedback with a message such as "Nice! Consistency. You Got This." to encourage the user to maintain concentration. The graphical visualization displays a slight decrease in focus score from approximately 85 percent to approximately 70 percent.

[0108] After approximately 35 minutes of study, the user moves away from the study area and the computer vision module detects that no face is present in the video frames. In step 406, the visualization and monitoring module determines that no face has been detected. In step 412, the dashboard interface generates a No Face Detected alert with a message such as "No Face Detected: Come Back to Screen" displayed within the system feedback section. The graphical visualization displays a sharp drop in focus score to approximately 0 percent during the period when no face is detected.

[0109] After the user returns to the study area, the computer vision module detects the user's face and resumes behavioral monitoring. However, the user exhibits behavioral indicators associated with reduced attention, including occasional off-screen gaze and increased blink rate. The artificial intelligence decision module classifies the user focus level as low focus. In step 412, the dashboard interface generates a Low Focus alert with a message such as "Stay Focused! Avoid Distractions." displayed within the system feedback section. In step 414, the system automatically increases LED brightness and shifts to a warm light tone to help the user recover concentration. In step 418, the Arduino microcontroller regulates the NeoPixel LED module to increase brightness to approximately 80 percent intensity with a color temperature of approximately 3000 Kelvin.

[0110] Throughout the study session, the method 400 continues to monitor the user's behavioral state and environmental conditions, generating appropriate alerts and feedback messages as conditions change. The graphical visualization on the dashboard interface displays the variations in focus score, room light percentage and camera brightness over time, enabling the user to observe how the system responds to behavioral and environmental changes. In step 420, the dashboard interface helps the user understand the system response by displaying feedback messages that explain the lighting adjustments made in response to detected changes in focus level and ambient illumination.

[0111] FIG. 5 illustrates a real-time monitoring dashboard interface displaying live system metrics according to various embodiments. The dashboard interface features a dark-themed graphical visualization panel positioned on the right side of the interface, showing a line graph titled "Live System Metrics" with three distinct colored traces representing different measured parameters over time. The horizontal axis represents time in seconds, ranging from 0 to 60, while the vertical axis displays percentage values from 0 to 100. A blue line represents the Room Light percentage, an orange line represents the Camera Brightness percentage, and a green line represents the Focus Score. In the visualization shown in FIG. 5, all three metrics appear relatively stable and horizontal across the displayed time period, with the Room Light percentage maintaining a value around 65 to 70 percent, the Camera Brightness percentage showing similar stable readings, and the Focus Score remaining consistent throughout the monitoring session. On the left side of the interface, a dark sidebar panel displays system status information including indicators for camera connection status and microcontroller connectivity. The interface may include control buttons for starting the camera and enabling alerts, along with a system status section showing connection states. The visualization module enables users to observe how the system continuously monitors environmental lighting conditions and user focus levels in real time, providing transparency into the adaptive lighting control process during study sessions.

[0112] FIG. 6 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a focus score variation according to various embodiments. The dashboard interface presents a line graph plotting three distinct parameters over time measured in seconds along the horizontal axis ranging from 0 to 60. The vertical axis displays

percentage values from 0 to 100. Three color-coded lines are shown: Room Light percentage represented in blue, Camera Brightness percentage shown in orange, and Focus Score displayed in green. The graph illustrates that the Room Light percentage and Camera Brightness percentage maintain relatively stable values around 60 to 70 percent throughout the monitoring period, while the Focus Score line shows a notable drop from approximately 100 percent to around 60 percent early in the session before stabilizing. The dashboard interface includes a navigation panel on the left side containing options including camera start controls, real-time alert toggles, and system status indicators showing microcontroller connectivity and camera active status elements. The visualization demonstrates the system's capability to continuously track and display environmental lighting conditions alongside computed user focus metrics, enabling real-time observation of how the system monitors both ambient illumination levels and user attention states during study sessions.

[0113] FIG. 7 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying detection of a sudden focus drop event according to various embodiments. The interface displays a graphical visualization titled "Live System Metrics" which displays three tracked parameters over time: Room Light percentage shown in blue, Camera Brightness percentage shown in orange, and Focus Score shown in green. The horizontal axis represents time in seconds ranging from 0 to approximately 60 seconds, while the vertical axis shows percentage values from 0 to 100. The graph demonstrates that the Room Light and Camera Brightness metrics remain relatively stable at approximately 60 to 70 percent throughout the monitoring period. However, the Focus Score line shows a dramatic and sudden drop from a stable level of approximately 60 to 70 percent down to nearly 0 percent at around the 55-second mark. The sharp decline in the focus score visually represents the system's capability to detect when a user's attention level suddenly decreases, which may trigger the adaptive lighting control mechanism to adjust the LED brightness and color temperature accordingly to help restore user concentration and reduce visual fatigue during study sessions. The left side of the interface displays a control panel showing system status information including camera activation controls and microcontroller connectivity indicators.

[0114] FIG. 8 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying fluctuations in room light and focus score according to various

embodiments. The dashboard presents a line graph with three distinct data traces plotted against time on the horizontal axis, which spans from 0 to approximately 60 units. The vertical axis represents percentage values ranging from 0 to 100. The blue line represents Room Light percentage, which shows fluctuations throughout the monitoring period, with values dropping sharply at multiple intervals to near zero before recovering to approximately 60 to 70 percent. The orange line represents Camera Brightness percentage, which maintains a relatively stable value around 60 to 65 percent throughout the displayed time period. The green line represents the Focus Score, which exhibits dramatic variations corresponding to changes in user attention, with values oscillating between approximately 20 percent and 100 percent. The graph demonstrates the system's capability to continuously monitor and visualize multiple parameters simultaneously, showing the correlation between environmental lighting conditions, camera-detected brightness levels and computed user focus scores. The left side panel of the interface shows system status indicators including camera start and alert enable options, along with system status information displaying microcontroller connected and camera active states. The visualization module enables users to observe how the system responds to behavioral and environmental changes in real time, providing transparency into the adaptive lighting control process.

[0115] FIG. 9 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a detected loss of user attention according to various embodiments. The dashboard interface features a dark-themed sidebar on the left side containing control elements including a camera start button, an alert enable toggle, and a system status section showing microcontroller connectivity with a connected indicator displayed in a highlighted box. The main portion of the interface presents a line graph titled "Live System Metrics" with the vertical axis scaled from 0 to 100 representing percentage values and the horizontal axis extending from 0 to approximately 60 representing time units. Three distinct metrics are plotted on the graph using different colored lines. The blue line represents Room Light percentage which maintains a relatively stable value around 80 to 85 throughout the monitoring period. The orange line represents Camera Brightness percentage which shows similar stability around 70 to 75 with a notable sharp dip occurring around the 35 to 40 time unit mark where the Camera Brightness percentage briefly drops to approximately 20 before recovering. The green line

represents Focus Score which demonstrates the variation, showing a dramatic drop from approximately 70 to near 0 around the 35 to 40 time unit mark, indicating a detected loss of user attention or focus during that period. The visualization demonstrates the system's capability to continuously monitor and graphically display environmental lighting conditions alongside computed focus scores, enabling real-time observation of how user attention levels fluctuate during study sessions and how the system tracks these changes for adaptive lighting control purposes.

[0116] FIG. 10 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying variations in system metrics over time according to various embodiments. The graph displays three distinct data traces plotted over time, with the horizontal axis representing elapsed time in seconds ranging from 0 to approximately 60 seconds, and the vertical axis showing percentage values ranging from 0 to 100. The blue trace represents Room Light percentage, which shows relatively stable values around 70 to 80 percent throughout the monitoring period, with a notable fluctuation occurring between approximately 20 and 35 seconds where the values temporarily increase toward 100 percent before returning to baseline levels. The green trace represents Camera Brightness percentage, which maintains relatively consistent values around 70 to 80 percent throughout the displayed time period. The yellow trace represents the Focus Score, which demonstrates variation during the monitoring session, showing a pronounced dip occurring between approximately 20 and 35 seconds where the focus score drops substantially before recovering to higher levels. The left side panel of the interface displays system status indicators including a camera start button, an alert enable toggle, and a system status section showing microcontroller connected and camera active status indicators, confirming active connectivity of the hardware components. The visualization demonstrates the system's capability to continuously monitor and display environmental lighting conditions, camera brightness levels, and computed user focus scores in real time, enabling observation of how these parameters fluctuate during study sessions and how the system tracks variations in user attention levels.

[0117] FIG. 11 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying system feedback messages during a study session according to various embodiments. The dashboard features a dark-themed graphical user interface with a

left-side control panel containing toggle options for camera start and alert enable along with a system status section showing microcontroller connected and camera active indicators. The main display area contains a line graph plotting three distinct parameters over time: Room Light percentage shown in blue, Camera Brightness percentage shown in orange, and Focus Score shown in green. The vertical axis of the graph ranges from 0 to 60, while the horizontal axis extends from 0 to approximately 60 time units. The plotted lines show relatively stable readings with the Room Light and Camera Brightness values maintaining consistent levels around 40 to 50 percent, while the Focus Score line remains steady at a lower value near the baseline. Below the graph, a system feedback section displays two notification messages: one indicating "You are in Deep Focus Mode" shown in a dark banner, and another stating "Room light is decreasing, LED brightness adjusted" displayed in a blue notification bar. The visualization demonstrates the system's capability to simultaneously monitor environmental lighting conditions, camera-detected brightness levels, and computed user focus scores while providing real-time adaptive feedback messages regarding the user's concentration state and automatic lighting adjustments made by the IoT-controlled LED system.

[0118] FIG. 12 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying motivational feedback and lighting adjustment notifications according to various embodiments. The dashboard features a dark-themed graphical user interface with a left-side control panel containing system status indicators including a camera start button, an alert enable toggle, and a system status section showing microcontroller connected and camera active status. The main central area displays a line graph plotting system metrics over time, with the vertical axis showing percentage values from 0 to 60 and the horizontal axis representing time intervals from 0 to approximately 60 units. The graph shows a green line that maintains a relatively stable level around 50 to 55 percent for the monitoring session, with a notable sharp dip occurring around the 30 to 35 time mark where the value drops to approximately 20 percent before recovering back to the baseline level. The dip pattern indicates a detected decrease in focus score or environmental lighting change during the monitoring period. Below the graph, a system feedback section displays two feedback messages: one indicating "Nice! Consistency. You Got This." as motivational feedback, and another stating "Room light

is decreasing. LED brightness adjusted." demonstrating the adaptive lighting control response to environmental changes. The upper right corner shows control buttons for managing the monitoring session.

[0119] FIG. 13 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying a no face detected alert according to various embodiments. The left side of the interface shows control elements including a camera start button, an alert enable toggle, and a system status section indicating microcontroller connected with a green indicator and camera active status. The central portion of the dashboard features a graphical visualization displaying three tracked parameters over time: Room Light percentage shown in blue, Camera Brightness percentage shown in orange, and Focus Score shown in green. The graph plots these values on a vertical axis ranging from 0 to approximately 60 to 80 units against a horizontal time axis extending from 0 to 60 seconds. The visualization shows relatively stable room light and camera brightness readings represented by horizontal lines in the upper portion of the graph, while the focus score line exhibits fluctuations, including notable drops to near-zero values around the 15 to 25 second mark, indicating periods where user attention decreased or face detection was lost. Below the graph, a system feedback section displays two alert messages: one stating "No Face Detected: Come Back to Screen" in a highlighted box, and another providing feedback about room lighting conditions. The dashboard demonstrates the system's capability to continuously monitor user focus levels and environmental conditions while providing real-time visual feedback and adaptive alerts to maintain study conditions.

[0120] FIG. 14 illustrates the real-time monitoring dashboard interface of FIG. 5 displaying distraction warning capabilities according to various embodiments. The interface is displayed within a web browser window and features a dark-themed design with a left sidebar containing system controls and status indicators. The sidebar includes options labeled camera start and alert enable along with a system status section showing connection indicators, with one appearing to show microcontroller connectivity status displayed in blue and another status indicator shown in a different color. The central portion of the dashboard displays a line graph visualization showing system metrics over time. The graph has a dark background with gridlines and displays data points connected by a green line. The vertical axis shows values ranging from

approximately 0 to 60, while the horizontal axis displays time intervals from 0 to approximately 60 units. The plotted line shows variations in the monitored parameter, with the values fluctuating between approximately 20 and 60 throughout the displayed time period. Notable dips in the graph indicate periods where the monitored metric decreased before recovering to higher levels. Below the graph is a system feedback section containing two feedback message boxes. The first message box displays a warning indicator with yellow coloring and contains text reading "Stay Focused! Avoid Distractions." indicating that the system has detected a potential distraction event. The second message box below displays information in blue text indicating that room light conditions are being monitored and LED brightness has been adjusted, demonstrating the adaptive lighting control functionality of the system responding to environmental conditions.

[0121] The dashboard interface visualizations shown in FIG. 5 through FIG. 14 demonstrate various operational states and system responses that may occur during study sessions. The graphical visualization component of the dashboard interface may plot multiple parameters simultaneously using color-coded line traces, enabling users to observe correlations between environmental lighting conditions, camera-detected brightness levels and computed focus scores. The system feedback section of the dashboard interface may display context-appropriate messages based on the current user state and environmental conditions, including Deep Focus Mode indicators, motivational feedback messages, no face detected alerts, distraction warnings and lighting adjustment notifications.

[0122] Working Example 3: Real-Time Dashboard Operation Across Different User States and Environmental Conditions

[0123] In a working example demonstrating real-time dashboard operation across different user states and environmental conditions, a user engages in an extended study session while the AI-driven adaptive smart lighting system monitors behavioral and environmental parameters. The visualization and monitoring module displays system metrics on the dashboard interface throughout the session, providing the user with real-time feedback regarding focus state and lighting adjustments.

[0124] At the beginning of the study session, the dashboard interface displays stable system metrics as illustrated in FIG. 5. The Room Light percentage maintains a

value around 65 to 70 percent, the Camera Brightness percentage shows similar stable readings around 65 to 70 percent, and the Focus Score remains consistent at approximately 80 to 85 percent. The system status indicators on the left sidebar show that the microcontroller is connected and the camera is active. The graphical visualization displays three horizontal traces indicating stable conditions across all monitored parameters.

[0125] After approximately 10 minutes of study, the user's attention begins to fluctuate as illustrated in FIG. 6. The Focus Score line on the graphical visualization shows a notable drop from approximately 100 percent to around 60 percent early in the monitoring period before stabilizing. The Room Light percentage and Camera Brightness percentage maintain relatively stable values around 60 to 70 percent throughout the monitoring period. The computer vision module detects the reduced gaze engagement and forwards the behavioral data to the artificial intelligence decision module for analysis.

[0126] At approximately 20 minutes into the study session, the user experiences a sudden loss of focus as illustrated in FIG. 7. The Focus Score line shows a dramatic and sudden drop from a stable level of approximately 60 to 70 percent down to nearly 0 percent at around the 55-second mark of the current monitoring window. The Room Light and Camera Brightness metrics remain relatively stable at approximately 60 to 70 percent throughout the monitoring period. The artificial intelligence decision module detects the sudden focus drop and may trigger the adaptive lighting control mechanism to adjust LED brightness and color temperature to help restore user concentration.

[0127] During the study session, environmental lighting conditions may fluctuate as illustrated in FIG. 8. The Room Light percentage shows fluctuations throughout the monitoring period, with values dropping sharply at multiple intervals to near zero before recovering to approximately 60 to 70 percent. The Focus Score exhibits dramatic variations corresponding to changes in user attention, with values oscillating between approximately 20 percent and 100 percent. The Camera Brightness percentage maintains a relatively stable value around 60 to 65 percent throughout the displayed time period. The system responds to the environmental lighting changes by adjusting LED brightness to compensate for the reduced ambient illumination.

[0128] At approximately 35 minutes into the study session, the dashboard interface displays a detected loss of user attention as illustrated in FIG. 9. The Camera Brightness percentage shows a notable sharp dip occurring around the 35 to 40 time unit mark where the Camera Brightness percentage briefly drops to approximately 20 before recovering. The Focus Score demonstrates a dramatic drop from approximately 70 to near 0 around the 35 to 40 time unit mark, indicating a detected loss of user attention or focus during that period. The Room Light percentage maintains a relatively stable value around 80 to 85 throughout the monitoring period.

[0129] The correlation between focus score variations and environmental changes is illustrated in FIG. 10. The Room Light percentage shows a notable fluctuation occurring between approximately 20 and 35 seconds where the values temporarily increase toward 100 percent before returning to baseline levels. The Focus Score demonstrates variation during the monitoring session, showing a pronounced dip occurring between approximately 20 and 35 seconds where the focus score drops substantially before recovering to higher levels. The Camera Brightness percentage maintains relatively consistent values around 70 to 80 percent throughout the displayed time period.

[0130] When the user maintains high focus during the study session, the dashboard interface displays Deep Focus Mode feedback and LED brightness adjustment notifications as illustrated in FIG. 11. The system feedback section displays a notification message indicating "You are in Deep Focus Mode" shown in a dark banner. A second notification states "Room light is decreasing, LED brightness adjusted" displayed in a blue notification bar, indicating that the system has automatically adjusted LED brightness in response to decreasing ambient light levels. The graphical visualization shows relatively stable readings with the Room Light and Camera Brightness values maintaining consistent levels around 40 to 50 percent.

[0131] When the user exhibits moderate focus levels, the dashboard interface provides motivational feedback as illustrated in FIG. 12. The system feedback section displays a message indicating "Nice! Consistency. You Got This." as motivational feedback to encourage the user to maintain concentration. A second message states "Room light is decreasing. LED brightness adjusted." demonstrating the adaptive lighting control response to environmental changes. The graphical visualization shows

a green line that maintains a relatively stable level around 50 to 55 percent for the monitoring session, with a notable sharp dip occurring around the 30 to 35 time mark where the value drops to approximately 20 percent before recovering back to the baseline level.

[0132] When the user moves away from the study area and the computer vision module detects that no face is present in the video frames, the dashboard interface generates a No Face Detected alert as illustrated in FIG. 13. The system feedback section displays an alert message stating "No Face Detected: Come Back to Screen" in a highlighted box. The graphical visualization shows the focus score line exhibiting fluctuations, including notable drops to near-zero values around the 15 to 25 second mark, indicating periods where user attention decreased or face detection was lost. The Room Light and Camera Brightness readings remain relatively stable in the upper portion of the graph.

[0133] When the user exhibits behavioral indicators associated with distraction, the dashboard interface displays distraction warning capabilities as illustrated in FIG. 14. The system feedback section displays a warning indicator with yellow coloring containing text reading "Stay Focused! Avoid Distractions." indicating that the system has detected a potential distraction event. A second message box displays information indicating that room light conditions are being monitored and LED brightness has been adjusted. The graphical visualization shows variations in the monitored parameter, with the values fluctuating between approximately 20 and 60 throughout the displayed time period, with notable dips indicating periods where the monitored metric decreased before recovering to higher levels.

[0134] Throughout the study session, the dashboard interface provides continuous real-time feedback that enables the user to observe how the adaptive lighting system responds to changes in behavioral state and environmental conditions. The graphical visualization component displays trends in focus score, ambient light percentage and camera brightness over time, while the system feedback section provides context-appropriate messages including Deep Focus Mode indicators, motivational feedback, no face detected alerts, distraction warnings and lighting adjustment notifications. The combination of graphical visualization and textual feedback enables users to

understand the relationship between their behavioral state, environmental lighting conditions and the adaptive lighting adjustments made by the system.

[0135] FIG. 15 illustrates a graph depicting adaptive lighting behavior based on user focus and environmental conditions over time according to various embodiments. The graph presents a line graph titled "Adaptive Lighting Behaviour Based on User Focus and Environmental Conditions" that illustrates the dynamic relationship between three system parameters over a fifty-minute time period. The vertical axis represents normalized values expressed as percentages ranging from approximately forty to eighty percent, while the horizontal axis displays time in minutes from zero to fifty minutes. Three distinct data series are plotted using different colored lines with circular markers at each data point.

[0136] A blue line represents the Focus Score, which begins at approximately eighty percent at time zero, decreases to around seventy-five percent at ten minutes, continues declining to approximately sixty-two percent at twenty minutes, reaches a low point of roughly fifty percent at thirty minutes, then recovers to approximately sixty-three percent at forty minutes before rising to around seventy percent at fifty minutes. An orange line depicts the Ambient Light Level, starting at approximately sixty percent, declining to around fifty-five percent at ten minutes, dropping further to approximately forty percent at twenty and thirty minutes, then increasing to roughly forty-seven percent at forty minutes and reaching approximately fifty percent at fifty minutes. A green line shows the LED Brightness Level, which begins at approximately fifty percent, rises to around fifty-five percent at ten minutes, increases to approximately sixty-five percent at twenty minutes, peaks at roughly seventy percent at thirty minutes, then decreases to approximately sixty percent at forty minutes and settles at around fifty-eight percent at fifty minutes.

[0137] The visualization depicted in FIG. 15 demonstrates the adaptive behavior of the AI-driven adaptive smart lighting system, showing how the LED brightness level inversely correlates with both the user focus score and ambient light conditions. When the focus score decreases from approximately eighty percent to fifty percent during the first thirty minutes, the LED brightness level correspondingly increases from approximately fifty percent to seventy percent. Similarly, when the ambient light level decreases from approximately sixty percent to forty percent, the LED brightness level

increases to compensate for the reduced environmental illumination. The inverse correlation between LED brightness and both focus score and ambient light level demonstrates the adaptive control mechanism that automatically increases illumination when focus decreases or ambient lighting diminishes.

[0138] Working Example 4: Experimental Validation of System Performance

[0139] To evaluate the effectiveness of the AI-driven adaptive smart lighting system, experimental testing was conducted under controlled indoor study conditions. The system setup consisted of a webcam for behavioral monitoring, a light dependent resistor sensor for detecting ambient light intensity, a microcontroller for lighting control and a programmable LED module. The computer vision component was developed using OpenCV and MediaPipe frameworks to detect facial landmarks and behavioral indicators such as eye movement, gaze direction and head posture, which were then used to estimate the user's level of attention during study sessions.

[0140] During the experimental evaluation, multiple study sessions were conducted in which the system continuously monitored both user behavior and environmental lighting conditions. The behavioral features extracted from the webcam were processed by the artificial intelligence decision module, which classified the user's attention level into three focus states: high focus, medium focus and low focus or fatigue. Based on the identified focus level, the system dynamically adjusted the brightness and color temperature of the LED lighting system.

[0141] Table 1 presents system performance metrics obtained during experimental validation testing.

Performance Metric	**Measured Value**
Focus Detection Accuracy	92.5%
Lighting Adjustment Response Time	Less than 500 milliseconds
Face Detection Rate	96.8%
False Positive Rate for Focus Classification	4.2%
System Latency (End-to-End)	Less than 800 milliseconds

[0142] Table 2 presents experimental data comparing system performance under various environmental lighting conditions.

Ambient Light Condition	Focus Detection Accuracy	LED Adjustment Accuracy	Response Time
Bright (greater than 70% ambient)	94.1%	95.3%	420 milliseconds
Moderate (40-70% ambient)	93.2%	94.8%	480 milliseconds
Dim (less than 40% ambient)	89.7%	93.1%	520 milliseconds

[0143] Table 3 presents experimental data comparing system performance across different user focus states.

Focus State	Classification Accuracy	Lighting Parameter Adjustment	User Concentration Improvement
High Focus	95.2%	Medium brightness, cool white	Baseline maintained
Medium Focus	91.8%	Balanced brightness, neutral white	12.4% improvement
Low Focus or Fatigue	90.1%	Increased brightness, warm tone	18.7% improvement

[0144] The experimental observations demonstrated that the system effectively responds to both behavioral and environmental changes. When stable gaze direction and minimal head movement were observed, the system recognized a high focus state and maintained steady cool-white lighting. When reduced attention, increased blinking, or irregular head movement was detected, the system adjusted lighting brightness and color temperature to create a more comfortable environment and reduce visual strain. The system automatically increases LED brightness when ambient light levels decrease or when the user's focus level drops, ensuring that appropriate lighting conditions are maintained throughout the study session.

[0145] Alternative Embodiments and Industrial Applicability

[0146] In some embodiments, the AI-driven adaptive smart lighting system may be implemented in educational institutions such as libraries, classrooms and study halls to support student concentration during learning activities. The system may be configured to monitor multiple users simultaneously by employing multiple webcams and processing video streams in parallel. In such configurations, the lighting control module may adjust lighting parameters for different zones within a room based on the aggregate focus levels of users in each zone.

[0147] In some embodiments, the system may be implemented in office environments to support worker productivity during extended work sessions. The artificial intelligence decision module may be configured with different focus thresholds and lighting parameters suited to office work activities. The system may integrate with building management systems to coordinate lighting control with other environmental parameters such as temperature and ventilation.

[0148] In some embodiments, the computer vision module may utilize alternative facial landmark detection algorithms or deep learning-based models for behavioral analysis. The system may employ convolutional neural network architectures trained on facial expression datasets to improve focus estimation accuracy. Machine learning-based decision logic may incorporate reinforcement learning techniques to adapt lighting parameters based on individual user preferences over time.

[0149] In some embodiments, the environmental sensing module may include additional sensors such as temperature sensors, humidity sensors, or noise level sensors to provide additional environmental context for lighting control decisions. The artificial intelligence decision module may analyze the combined sensor data to determine lighting parameters that account for multiple environmental factors affecting user comfort and concentration.

[0150] In some embodiments, the visualization and monitoring module may be implemented as a mobile application that enables users to monitor system status and receive alerts on portable devices. The dashboard interface may provide historical analytics showing focus patterns and lighting adjustments over extended time periods. The system may generate reports summarizing user focus levels and study session productivity metrics.

[0151] In some embodiments, the programmable LED lighting system may include multiple lighting fixtures distributed throughout a room to provide zone-based lighting control. The microcontroller may control individual lighting fixtures independently to create localized lighting environments suited to different activities or user positions within the room. The system may employ wireless communication protocols to enable flexible placement of lighting fixtures without requiring wired connections.

[0152] The AI-driven adaptive smart lighting system provides technical effects including non-intrusive monitoring of user cognitive state without requiring wearable physiological sensors, thereby reducing system complexity and improving practicality for everyday usage. The integration of behavioral monitoring with environmental sensing allows the system to make context-aware lighting adjustments that respond to both user attention levels and surrounding illumination conditions simultaneously. The continuous monitoring loop enables the system to dynamically adapt lighting conditions in response to changes in user attention and ambient illumination throughout study sessions without requiring manual adjustment. The differentiated lighting responses for each focus state enable the system to provide appropriate visual environments tailored to the user's cognitive condition, where cool white lighting during high focus states supports alertness and concentration, while warm light tones during low focus or fatigue states reduce eye strain and improve visual comfort.

[0153] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field.

[0154] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in

the art and are intended to fall within the scope of the invention as defined in the following claims.

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WE CLAIM:

1. An artificial intelligence-driven adaptive lighting system (100), comprising:

a computer vision module configured to capture real-time video frames of a user via a camera (102) and process the video frames using computer vision frameworks (104) to detect facial landmarks, gaze direction and head orientation (106);

an environmental sensing module having a light dependent resistor sensor connected to a microcontroller, the environmental sensing module configured to measure ambient light intensity within an environment (108);

an artificial intelligence decision module configured to:

receive behavioral data from the computer vision module and environmental data from the environmental sensing module (302);

analyze the behavioral data and the environmental data using focus thresholds and machine learning-based decision logic (304); and

classify a user focus level into a high focus state, a medium focus state, or a low focus state (306); and

an Internet of Things-enabled lighting control module having a programmable LED lighting system connected to the microcontroller, the lighting control module configured to receive lighting parameters from the artificial intelligence decision module (314) and adjust brightness and color temperature of the programmable LED lighting system based on the classified user focus level (316).

2. The system of claim 1, wherein the computer vision module is configured to monitor behavioral indicators including eye blink rate, gaze direction, head orientation and facial posture to estimate user attention levels (210).

3. The system of claim 1, wherein the artificial intelligence decision module is configured to:

maintain steady medium brightness with cool white light when the user focus level is classified as the high focus state (308);

apply balanced brightness with neutral white light when the user focus level is classified as the medium focus state (310); and

increase brightness and shift to a warm light tone when the user focus level is classified as the low focus state (312).

4. The system of claim 1, wherein the computer vision module is configured to generate a no face detected alert when no face is detected in the video frames (208).

5. The system of claim 1, wherein the lighting control module is configured to transmit control signals to the programmable LED lighting system via a serial communication interface (314).

6. The system of claim 1, wherein the system is configured to automatically increase LED brightness when ambient light levels decrease or when the user focus level drops (414).

7. The system of claim 1, further comprising a visualization and monitoring module configured to provide a dashboard interface (402) that displays focus score, ambient light percentage, camera brightness, session duration and system alerts (404).

8. The system of claim 7, wherein the dashboard interface is configured to:

display a deep focus mode indicator when the user is in the high focus state (408);

provide motivational feedback when the user is in the medium focus state (410); and

generate a low focus alert when attention loss is detected (412).

9. The system of claim 1, wherein the computer vision module is configured to convert the behavioral indicators into structured data representing a user state (214) and forward the structured data to the artificial intelligence decision module (216).

10. The system of claim 1, wherein the artificial intelligence decision module is configured to determine whether conditions have changed (318) and maintain current lighting conditions when conditions have not changed (320).

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ABSTRACT

ADAPTIVE LIGHTING SYSTEM WITH COMPUTER VISION-BASED FOCUS DETECTION

An artificial intelligence-driven adaptive lighting system (100) comprises a computer vision module configured to capture real-time video frames of a user via a camera (102), process the video frames using computer vision frameworks (104), and detect facial landmarks, gaze direction and head orientation (106). The system comprises an environmental sensing module having a light dependent resistor sensor connected to a microcontroller configured to measure ambient light intensity (108). The system comprises an artificial intelligence decision module configured to receive behavioral and environmental data (302), analyze the data using focus thresholds and machine learning-based decision logic (304), and classify a user focus level into a high focus state, a medium focus state, or a low focus state (306). The system comprises an Internet of Things-enabled lighting control module configured to adjust brightness and color temperature of a LED lighting system based on the classified focus level.

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